

In our quest for tree-decompositions into ‘small’ bags, we now have two criteria to choose between: the global ‘worst case’ criterion of width, and the more subtle local criterion of leanness. Surprisingly, it is always possible to find a tree-decomposition that is optimal with respect to both criteria at once:

Theorem 12.4.5. (Thomas 1990)

Every graph G has a lean tree-decomposition of width $\text{tw}(G)$.

Another natural feature one might ask of a tree-decomposition is that its parts, as induced subgraphs of G , be connected. Let us call such a tree-decomposition *connected*, and let the *connected tree-width* $\text{ctw}(G)$ of G be the least width of a connected tree-decomposition of G .

The connected tree-width of most graphs is greater than their ordinary tree-width. For example, every cycle has tree-width 2, but $\text{ctw}(C^n) = \lceil n/2 \rceil$ (Exercise 33). And unlike ordinary tree-width, the connected tree-width of a subgraph of G can be greater than that of G . (For example, let G be obtained from a long cycle by adding a chord between opposite vertices.) However, if $C \subseteq G$ is a *geodesic* cycle, i.e., if $d_C(u, v) = d_G(u, v)$ for all vertices $u, v \in C$, then $\text{ctw}(C) \leq \text{ctw}(G)$.

The presence of long geodesic cycles in a graph thus is an obstruction to small connected tree-width – as is, trivially, large ordinary tree-width. By the following theorem, however, these are the only obstructions:

Theorem 12.4.6. *There is a function $f: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that every graph of tree-width $\leq w \in \mathbb{N}$ that has no geodesic cycle of length $> k \in \mathbb{N}$ has connected tree-width at most $f(w, k)$.*

12.5 Tangles

We have already in this chapter met a few types of substructures of possibly sparse graphs that are highly connected in some sense, but which are not just k -connected subgraphs (or minors) for some large k . Large grid minors are an example, the k -blocks defined at the end of Section 12.3 are another, as are brambles of high order. All these substructures have one feature in common: for every low-order separation of the graph they lie essentially, though not necessarily entirely, on one of its two sides.

For example, given a bramble $\mathcal{B}$ of order k in a graph $G = (V, E)$, if $\{A, B\}$ is a separation of order $< k$ then $A \setminus B$ or $B \setminus A$, but not both, contains a set from $\mathcal{B}$. This helped us prove the easy implication of the tree-width duality theorem: $\mathcal{B}$ orients every edge of the decomposition tree T of any tree-decomposition $(T, \mathcal{V})$ of adhesion $< k$ ‘towards’ the

side of its induced separation that contains one of the sets in $\mathcal{B}$, and the edges thus oriented point to a central node t of T for which V_t covers $\mathcal{B}$.

It has turned out that, more often than not, the only feature of a highly connected substructure that mattered in the proofs of theorems about it was the information of how it ‘orients’ the low-order separations of G in this way. Collecting just this information together leads to a more abstract notion of a highly connected substructure, called a *tangle*. The purpose of this section is to make this precise, to prove a duality theorem for tangles in the spirit of Theorem 12.4.3, and to point out how this setting can be used to express the duality between global tree-structure of a graph and its highly connected substructures more generally.

In the context of tangles, we often denote the order $|A \cap B|$ of a separation $s = \{A, B\}$ simply as $|s|$. The *orientations* of $\{A, B\}$ are the two *oriented separations* (A, B) and (B, A) . We say that (A, B) is oriented, or ‘points’, *towards* B and its subsets. Given oriented separations (A, B) and (C, D) , we write $(A, B) \geq (C, D)$ if $A \subseteq C$ and $B \supseteq D$.

Two separations of G are *nested* if they have $\leq$ -comparable orientations; otherwise they *cross*. The separation $\{E, F\}$ in Figure 12.5.1, for example, is nested with the two crossing separations $\{A, B\}$ and $\{C, D\}$, because $(E, F) \geq (A, B), (C, D)$. The separations induced by a tree-decomposition are clearly nested, and every nested set of separations of a graph is induced by some tree-decomposition; see after Lemma 12.3.1.

Given two separations $\{A, B\}$ and $\{C, D\}$ of G , it is easy to check that also $\{A \cup C, B \cap D\}$ is a separation of G . We call the four separations of this form the *corners* of the separations $\{A, B\}$ and $\{C, D\}$ (Fig. 12.5.1). They come in pairs of *opposite* corners: the corner opposite to $\{A \cup D, B \cap C\}$, for example, is $\{B \cup C, A \cap D\}$. Two corners that are not opposite are *adjacent*. The adjacent corners $\{A \cup C, B \cap D\}$ and $\{A \cup D, B \cap C\}$ lie on the same side of $\{A, B\}$, on its *B-side*.

 $|s|$ oriented
separation $\leq$

nested

cross

corners

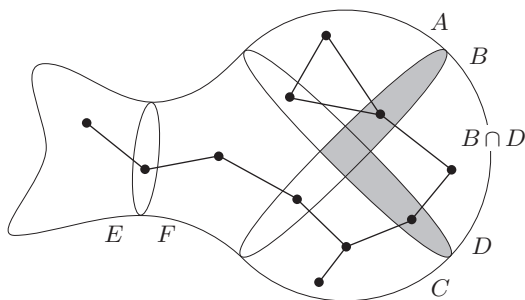
opposite
adjacent

Fig. 12.5.1. The separation $\{A \cup C, B \cap D\}$ is one of the four corners of $\{A, B\}$ and $\{C, D\}$. The separation $\{E, F\}$ is nested with $\{A, B\}$ and $\{C, D\}$ and all their corners.

The sum of the orders of a pair of opposite corners always equals the sum of the orders of the original two crossing separations. For example:

$$|(A \cup D) \cap (B \cap C)| + |(B \cup C) \cap (A \cap D)| = |A \cap B| + |C \cap D|.$$

The $\leq$ -part of this equality, the fact that opposite corners c, d of any two separations r, s satisfy

$$\text{submodularity} \quad |c| + |d| \leq |r| + |s|, \quad (\dagger)$$

is sometimes referred to as *submodularity* of the order function $s \mapsto |s|$. It implies, for example, that of any two opposite corners of two separations of order $< k$ at least one also has order $< k$. More generally, $(\dagger)$ implies that $|c| \leq |r|$ or $|d| \leq |s|$ (or both), which is the standard way to apply it.

$\vec{S}$ Given a set S of separations, we write $\vec{S} := \{(A, B) \mid \{A, B\} \in S\}$ for the set of all their orientations. An *orientation* of S is a subset of $\vec{S}$ that contains for every element of S exactly one of its two orientations. Unless otherwise mentioned, from now on in this section S will be the set of all the separations of G , so $\vec{S}$ is the set of all its oriented separations.

S A set σ of oriented separations of G is *consistent* if it does not contain (B, A) whenever $(A, B) \geq (C, D)$ with $(C, D) \in \sigma$.⁵

consistent For example, if $(T, \mathcal{V})$ is a tree-decomposition of G with $\mathcal{V} = (V_t)_{t \in T}$, and V_t is not contained in any other bag, then orienting the separations that $(T, \mathcal{V})$ induces towards V_t orients them consistently. A k -block X of G even defines a consistent orientation of the entire set

$$S_k := \{s \in S : |s| < k\}$$

of separations of order $< k$ of G : the orientation $\{(A, B) \in \vec{S}_k \mid X \subseteq B\}$.

A *bramble* $\mathcal{B}$ of order $n \geq k$ also defines an orientation of S_k : the set $O = \{(A, B) \in \vec{S}_k \mid \exists X \in \mathcal{B} : X \subseteq B \setminus A\}$. Unlike in the case of k -blocks, there need not be any fixed bramble set that lies in every B with $(A, B) \in O$; indeed the intersection of all these B may be empty. (Example?) But O then shows the large order of $\mathcal{B}$ in another way, in that we cannot cover $\mathcal{B}$ by few sets A with $(A, B) \in O$: since any bramble set meeting A also meets $A \cap B$ (as it touches the bramble set in $B \setminus A$), and $|A \cap B| \leq k - 1$, we need at least $n/(k - 1)$ sets A to cover $\mathcal{B}$. The idea that this is enough not only to reflect but to constitute a kind of highly connected substructure in G has led to the following concept of a tangle.

avoids An orientation O of a set of separations *avoids* a set $\mathcal{F}$ of subsets of $\vec{S}$ if no subset of O is an element of $\mathcal{F}$. A *tangle of order k* in G , or *k-tangle* for short, is an orientation of S_k that avoids

$$\mathcal{T} := \{ \{(A_1, B_1), (A_2, B_2), (A_3, B_3)\} \mid G[A_1] \cup G[A_2] \cup G[A_3] = G \}.$$

⁵ Intuitively, σ is consistent if no two of its elements point away from each other. In particular, it will not contain both orientations of any given separation.

Note that the separations (A_i, B_i) in the definition of $\mathcal{T}$ need not be distinct. For example, no tangle contains a separation of the form (V, A) . And if $(A, B) \geq (C, D)$ then no tangle contains both (B, A) and (C, D) , since $G[B] \cup G[C] \supseteq G[D] \cup G[C] = G$. Thus, tangles are consistent. More generally, tangles τ clearly have the following two properties:

If τ contains the oriented separations (A, B) and (C, D) ,
it does not contain their oriented corner $(B \cap D, A \cup C)$. (P)

Given separations $\{A, B\}$ and $\{C, D\}$, if τ contains (A, B)
it does not contain both $(B \cap D, A \cup C)$ and $(B \cap C, A \cup D)$. (R)

Sets of oriented separations satisfying (P) are known as *profiles*, and (P) is the *profile property* of tangles.⁶ Note that it can be satisfied in two ways: either by $(A \cup C, B \cap D) \in \tau$, or by the fact that τ does not orient the corner $\{A \cup C, B \cap D\}$ at all; this happens if that corner has order k or larger, but τ is only a k -tangle. Profiles satisfying (R) are *robust*.

profile

Like its k -blocks, compare Theorem 12.3.7, the tangles of a graph can be ‘separated’ by a tree-decomposition. Since the tangles in a graph see it in terms of its separations rather than its subgraphs, this is naturally expressed not in terms of the bags of the decomposition but of the separations it induces.

Let us say that a separation *distinguishes* two tangles, not necessarily of the same order, if they both orient it but do so differently. Distinct tangles of the same order are always distinguished by some separation; otherwise they would be identical. But a k -tangle cannot be distinguished from the ℓ -tangle it *induces* for $\ell < k$, the set of those of its oriented separations that have order $< \ell$.

distinguish

*induced
tangle*

A separation that distinguishes two tangles does so *efficiently* if they are not distinguished by any separation of smaller order. A set T of separations *distinguishes* some set of tangles in a graph G *efficiently* if every two tangles in this set that are distinguished by some separation of G are distinguished efficiently by a separation in T .

efficient

Theorem 12.5.1. (Robertson & Seymour 1991)

Every graph G has a nested set of separations that distinguishes all the tangles in G efficiently.

*tree-of-
tangles
theorem*

It is not hard to construct from any nested set of separations, such as that in Theorem 12.5.1, a tree-decomposition that induces precisely this set of separations as defined after Lemma 12.3.1 (Exercises 16 and 51).

⁶ If we think of tangles as orienting separations towards their ‘big’ side, condition (P) is reminiscent of ultrafilters: the intersection of two big sets cannot be small.

We shall give two proofs for Theorem 12.5.1. The first proof is based on a combinatorial lemma, the *splinter lemma*, which encapsulates the essence of earlier direct proofs that sought to find the required nested set of tangle-distinguishing separations directly among the corners of arbitrary efficient tangle-distinguishing separations.

Our second proof of Theorem 12.5.1 comes with an important strengthening: the nested set of tangle-distinguishing separations it finds is not constructed by a sequence of choices, as in the first proof, but is identified by an explicitly stated property of separations. As a consequence, this set is invariant under the automorphisms of the graph G . We state this result separately as Theorem 12.5.8.

In both proofs we shall need the following lemma about nested and crossing separations (see [Figure 12.5.1](#)):

Lemma 12.5.2. (Fish Lemma)

Any separation r that is nested with two crossing separations s and t is also nested with their four corners.

Proof. Since r is nested with s , the two have orientations $\vec{r} \geq \vec{s}$ (Ex. 40). Similarly, r and t have orientations $\vec{r}' \geq \vec{t}'$. If $\vec{r}' \neq \vec{r}$, then $\vec{r} \leq \vec{t}$ with $\vec{t}$ the inverse of $\vec{t}'$. Then $\vec{s} \leq \vec{r} \leq \vec{t}$, showing that s and t are nested. This contradicts our assumptions, so $\vec{r}' = \vec{r} \geq \vec{s}, \vec{t}'$.

Let $\vec{r} =: (E, F)$ and $\vec{s} =: (A, B)$ and $\vec{t}' =: (C, D)$ ([Fig. 12.5.1](#)). Then $(E, F) \geq (A \cap C, B \cup D)$ as well as, more trivially, $(E, F) \geq (A \cup C, B \cap D)$ and $(E, F) \geq (A \cup D, B \cap C')$ and $(E, F) \geq (C \cup B, A \cap D)$. In particular, r is nested with all four corners of s and t . $\square$

Let us say that a family $(A_1, \dots, A_n)$ of sets of separations of a graph *splinters* if, for all $i \neq j$, any crossing $a_i \in A_i \setminus A_j$ and $a_j \in A_j \setminus A_i$ have a corner in $A_i \cup A_j$.

Lemma 12.5.3. (Splinter Lemma)

Every splintering family $(A_1, \dots, A_n)$ of non-empty sets of separations has a nested family $(a_1, \dots, a_n)$ of representatives $a_i \in A_i$.

Proof. We apply induction on n , which starts trivially with $n = 1$. For the induction step let $(a_1, \dots, a_{n-1})$ be a nested family of representatives $a_i \in A_i$ of $(A_1, \dots, A_{n-1})$. If any of those a_i lies in A_n , let $a_n := a_i$; then $(a_1, \dots, a_n)$ is as desired. Suppose now that $a_i \in A_i \setminus A_n$ for all $i < n$, and choose $a_n \in A_n$ so as to cross a_i for as few $i < n$ as possible.

Let us show that, for all $i < n$, the set A'_i of elements of A_i that are nested with a_n is non-empty. If $a_n \in A_i$ then $a_n \in A'_i$; we may thus assume that $a_n \in A_n \setminus A_i$ (as well as $a_i \in A_i \setminus A_n$). We may also assume that a_i and a_n cross, as otherwise $a_i \in A'_i$. Since $(A_1, \dots, A_n)$ splinters, a_i and a_n have a corner c in $A_i \cup A_n$. By the fish lemma, c is nested with all the a_j that a_n is nested with (since these a_j are also nested with a_i),

and with a_i in addition. Hence $c \notin A_n$ by the choice of a_n , and therefore $c \in A_i$. Since c is nested with a_n , this yields $c \in A'_i$. This completes the proof that the A'_i are non-empty.

Let us show that the family $(A'_1, \dots, A'_{n-1})$ splinters. Consider any crossing separations $a'_i \in A'_i \setminus A'_j$ and $a'_j \in A'_j \setminus A'_i$ with $i \neq j$. Since a'_i is nested with a_n it cannot lie in $A_j \setminus A'_j$, and similarly $a'_j \notin A_i \setminus A'_i$. Thus, $a'_i \in A_i \setminus A_j$ and $a'_j \in A_j \setminus A_i$. As $(A_1, \dots, A_n)$ splinters, a'_i and a'_j have a corner c in $A_i \cup A_j$. We need to show that $c \in A'_i \cup A'_j$, i.e., that c is nested with a_n . But this follows from the fish lemma, since a_n is nested with both $a'_i \in A'_i$ and $a'_j \in A'_j$.

As the family $(A'_1, \dots, A'_{n-1})$ splinters, the induction hypothesis implies that it has nested representatives $a''_i \in A'_i$. As these are nested with a_n by definition of the A'_i , the family $(a''_1, \dots, a''_{n-1}, a_n)$ can serve as our desired nested family of representatives of $(A_1, \dots, A_n)$. $\square$

Proof of Theorem 12.5.1. Pick an enumeration of the distinguishable pairs of tangles in G , and let A_i denote the set of separations of G that efficiently distinguish the i th pair. By Lemma 12.5.3 it suffices to show that the family of these A_i splinters.

To prove this, consider any indices $i \neq j$ with crossing separations $s \in A_i \setminus A_j$ and $t \in A_j \setminus A_i$. Assume that these have orders $|s| \leq |t|$; that s efficiently distinguishes the i th pair of tangles, σ and σ' ; and that t efficiently distinguishes the j th pair of tangles, τ and τ' (Figure 12.5.2). As $|s| \leq |t|$, the tangle τ orients s ; we may assume it orients s as σ does. As s is not in A_j but has order $|s| \leq |t|$, it does not distinguish τ from τ' ; so τ' too orients s as τ and σ do.

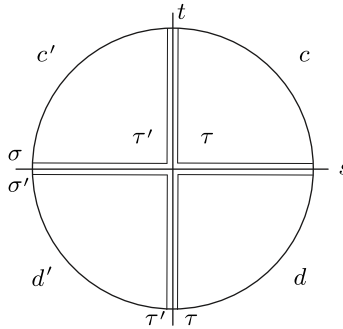


Fig. 12.5.2. Crossing separations $s \in A_i \setminus A_j$ and $t \in A_j \setminus A_i$ and their four corner separations c, c', d, d'

Consider the corner c of s and t at the top right in Figure 12.5.2. If c has order $|c| \leq |t|$, then τ orients it. Since τ has the profile property (P), and it orients t towards the right in the diagram and s upwards, it cannot orient c towards the centre. So τ orients c towards the top right. But τ' orients c towards the centre, since it orients t towards the left and

is consistent. Thus, c distinguishes τ from τ' , and by our assumption that $|c| \leq |t|$ it does so efficiently, giving $c \in A_j$ as desired. We may thus assume that $|c| > |t|$. Analogously with c' instead of c , we are done unless $|c'| > |t|$, which we therefore assume.

But now, by submodularity ($\dagger$), these assumptions about the orders of c and c' imply that their opposite corners d' and d have order $< |s|$. Hence both σ and σ' orient them. As σ orients s upwards, it must orient d and d' towards the centre, by consistency. But since d and d' have order $< |s|$, they cannot distinguish σ from σ' , so σ' too orients them towards the centre. As σ' orients s downwards, it thus violates (R).

This contradiction completes the proof that the family of all A_i splinters, as was our aim to show. $\square$

The construction of a nested set of efficient tangle-distinguishing separations in our first proof of Theorem 12.5.1 makes a number of choices, those of the a_n at the start of the proof of the splinter lemma. Objects defined for a graph without making such choices, but so that they are invariant under its automorphisms,⁷ are called *canonical*. Our second proof of Theorem 12.5.1 provides such a canonical nested set of tangle-distinguishing separations: the set of *all separations that efficiently distinguish some pair of tangles and cross as few other such separations as possible*. We shall make this definition precise later.

Given a pair τ, τ' of tangles in G , let $D(\tau, \tau')$ denote the set of separations of G that distinguish τ, τ' efficiently. Our second proof of Theorem 12.5.1 rests on a property of these sets that has its own name. An *entanglement* in G is any non-empty set T of separations of G that satisfies the following condition:

Whenever some $t \in T$ is crossed by a separation s of G so that two corners of s and t that lie on the same side of t have order at most $|t|$, then at least one of these corners has order exactly $|t|$ and also lies in T . (E)

Lemma 12.5.4. *For every pair τ, τ' of distinguishable tangles in G , the separations of G that distinguish them efficiently form an entanglement.*

Proof. Let crossing separations $t \in D(\tau, \tau')$ and s be given as in condition (E), with corners c, d on the same side of t . These are as shown in Figure 12.5.2, together with the tangles τ, τ' that t distinguishes.

Since c and d have order at most $|t|$, the tangle τ orients them. As τ orients t towards the right, it cannot orient both c and d inwards, by (R). But if τ orients c outwards, say, then c distinguishes τ from τ' , which orients t towards the left and hence, by consistency, orients c inwards. Since t distinguishes τ and τ' efficiently we cannot have $|c| < |t|$, so $|c| = |t|$ as required. $\square$

⁷ See the discussion following Theorem 12.3.7 for why this can be important.

We need three more general lemmas about crossing separations.

Lemma 12.5.5. *Let s, t be two crossing separations of G , with opposite corners c, d . Then every separation r of G satisfies the following:*

- (i) *If r crosses c or d , it also crosses s or t .*
- (ii) *If r crosses c and d , it also crosses s and t .*

Proof. (i) is equivalent to the fish lemma (12.5.2).

(ii) If $r = \{A, B\}$ and $\{C, D\} \in \{s, t\}$ are nested, they have comparable orientations; indeed we may assume that $(A, B) \geq (C, D)$ (Ex. 40). Since c and d are opposite corners of s and t , one of them lies on the D -side of $\{C, D\}$. This corner has a side contained in $D \subseteq B$, while its other side contains $C \supseteq A$. It is therefore nested with r . $\square$

Given a set S of separations of G and any separation a of G , let $x_S(a) =: x(a)$ denote the number of separations in S that a crosses.

S
 x

Lemma 12.5.6. *Let $s, t \in S$ be two crossing separations, with opposite corners c, d . Then $x(c) + x(d) < x(s) + x(t)$.*

Proof. To evaluate x on our separations c, d, s, t , we have to count the separations $r \in S$ which they cross. Counting only separations $r \notin \{s, t\}$, we obtain $x(c) + x(d) \leq x(s) + x(t)$ from Lemma 12.5.5. The inequality gets strict when we allow $r \in \{s, t\}$, since s and t cross each other but not their corners c and d . $\square$

Let $\mathcal{S}$ be any set of sets of separations of G .⁸ Let us say that a separation s of G is $\mathcal{S}$ -friendly if it lies in some $S \in \mathcal{S}$ such that no other separation in S crosses fewer separations in $\bigcup \mathcal{S}$ than s does. Note that every $S \in \mathcal{S}$ has at least one $\mathcal{S}$ -friendly element.

$\mathcal{S}$ -friendly

Lemma 12.5.7. *Suppose that for all sets $S, S' \in \mathcal{S}$, possibly $S = S'$, and any crossing separations $s \in S$ and $s' \in S'$, one of the following two assertions holds:*

- (i) *s and s' have opposite corners $c \in S$ and $c' \in S'$;*
- (ii) *s and s' have pairs of opposite corners c, d in S and c', d' in S' .*

Then the $\mathcal{S}$ -friendly separations of G are nested.

Proof. Let us show that crossing separations s, s' of G cannot both be $\mathcal{S}$ -friendly. Suppose they are. Then $s \in S$ and $s' \in S'$ for some $S, S' \in \mathcal{S}$ witnessing this. By assumption, these satisfy (i) or (ii).

Suppose first that (i) holds. Then $x(c) + x(c') < x(s) + x(s')$ by Lemma 12.5.6, where $x = x_{\hat{S}}$ for $\hat{S} = \bigcup \mathcal{S}$. So either $x(c) < x(s)$ or $x(c') < x(s')$. In the first case s is not $\mathcal{S}$ -friendly, by $c \in S$ and our choice of S for s . In the second case s' is not $\mathcal{S}$ -friendly, by $c' \in S'$ and our choice of S' for s' .

⁸ When we apply Lemma 12.5.7 later, $\mathcal{S}$ will be a set of entanglements in G .